GIFT4U

Wedding Marketplace

Flowcharts + Entity Relationship Diagrams

SYSTEM DESIGN PACKAGE

India-focused wedding shopping architecture covering communities, states, ceremonies, product categories, verified ceremony needs, seller SKUs, customizable combos, payment and fulfilment.

31	316	307	09
Community templates	Ceremony slots	Product types	System diagrams

HOW TO READ

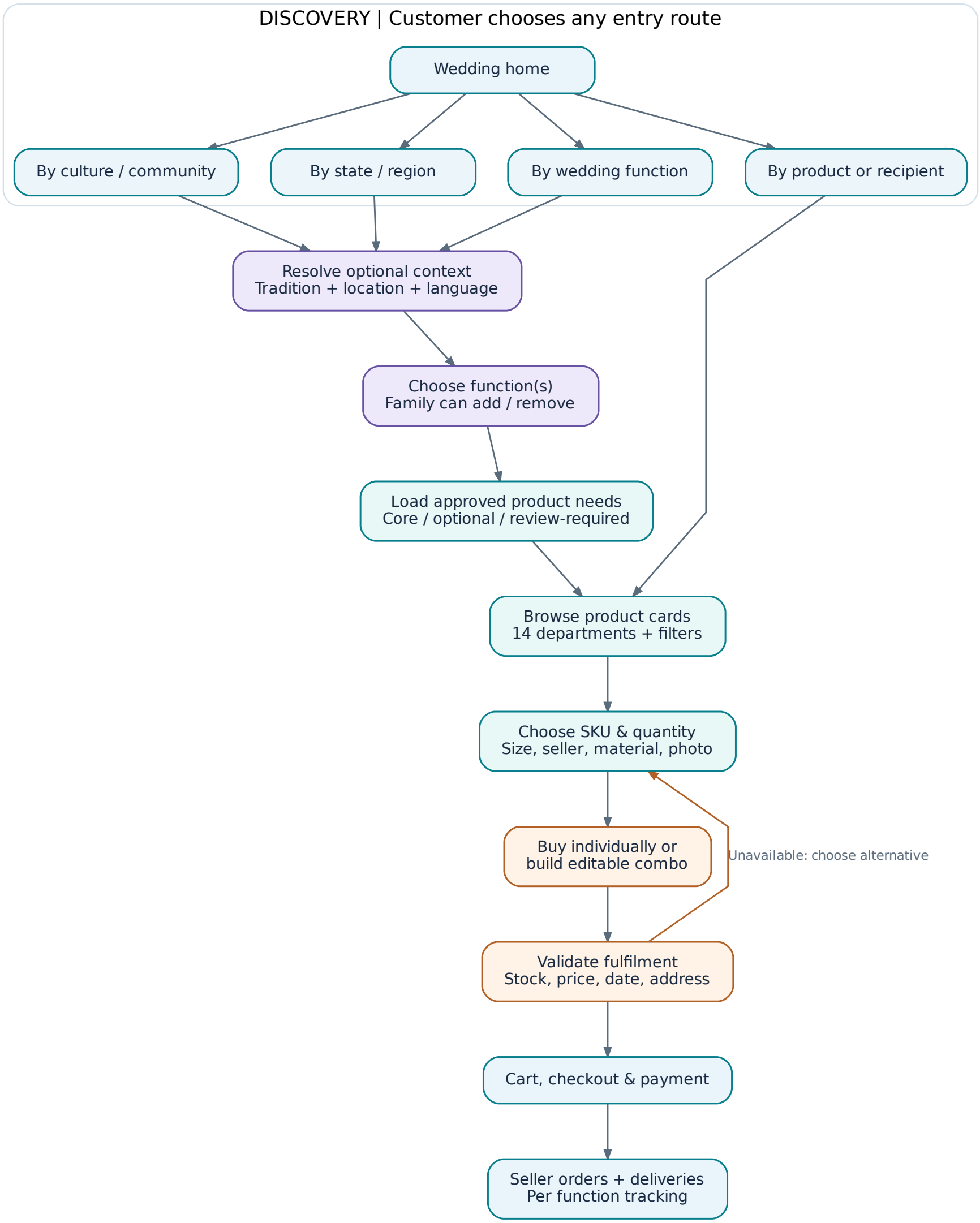
Master records · Join / configuration · Product / active choice · Validation / stock

Core design rules

1. Canonical products & SKU inventory - never duplicate stock per culture.
2. Ritual needs are approved per tradition + ceremony, not auto-inferred from state.
3. Combo blueprints are reusable; buyer selections and placed orders are snapshots.

Customer shopping journey

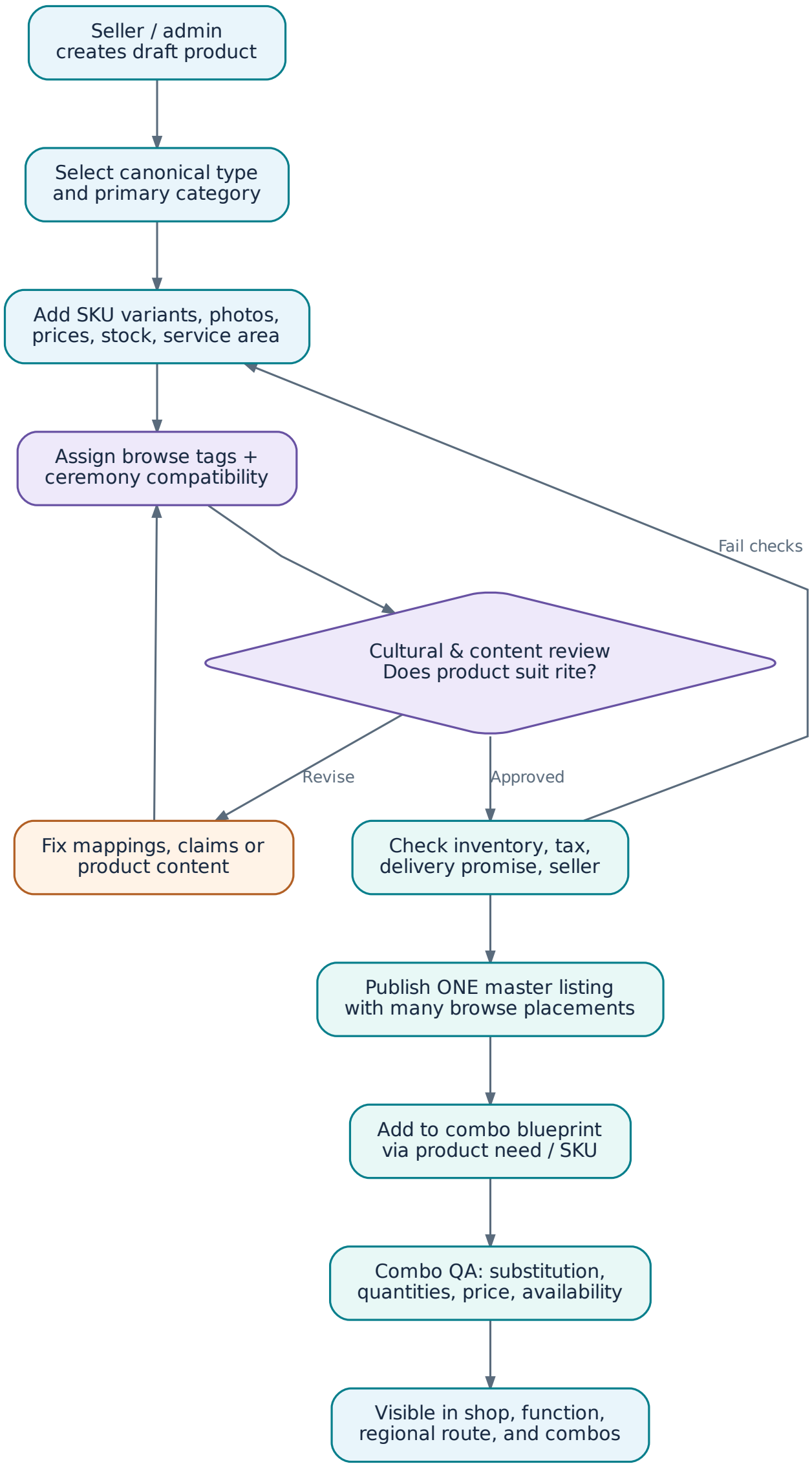
Multiple ways to browse lead to the same approved product catalogue and one checkout.



Do not force religion selection. Validate prices, sellers, SKU stock and delivery before accepting orders.

Admin listing & combo workflow

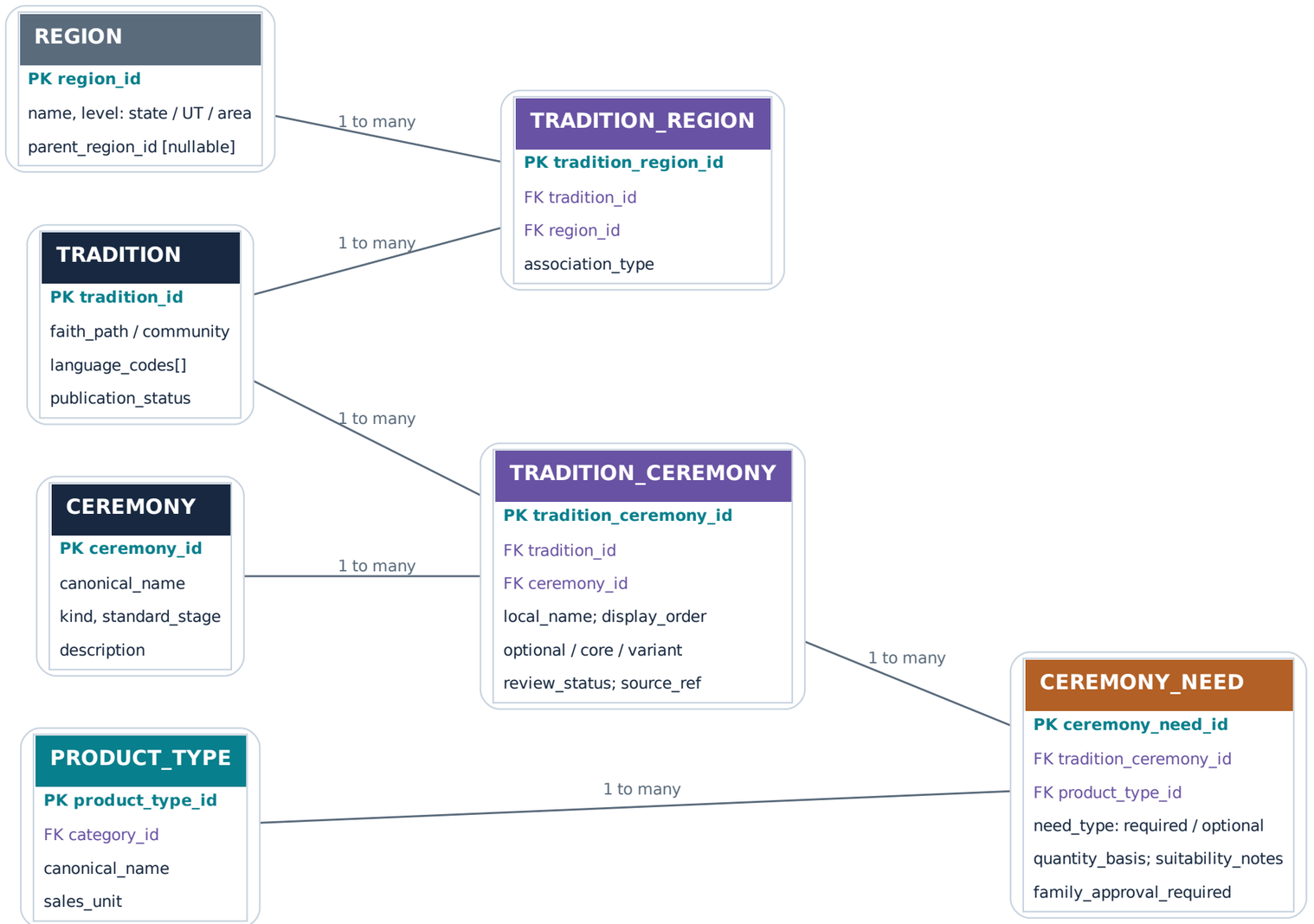
Draft -> classify -> culturally review -> validate fulfilment -> publish -> connect combos.



A product can be present in many browse collections but has only one canonical listing per seller.

ER: Culture, region & ceremonies

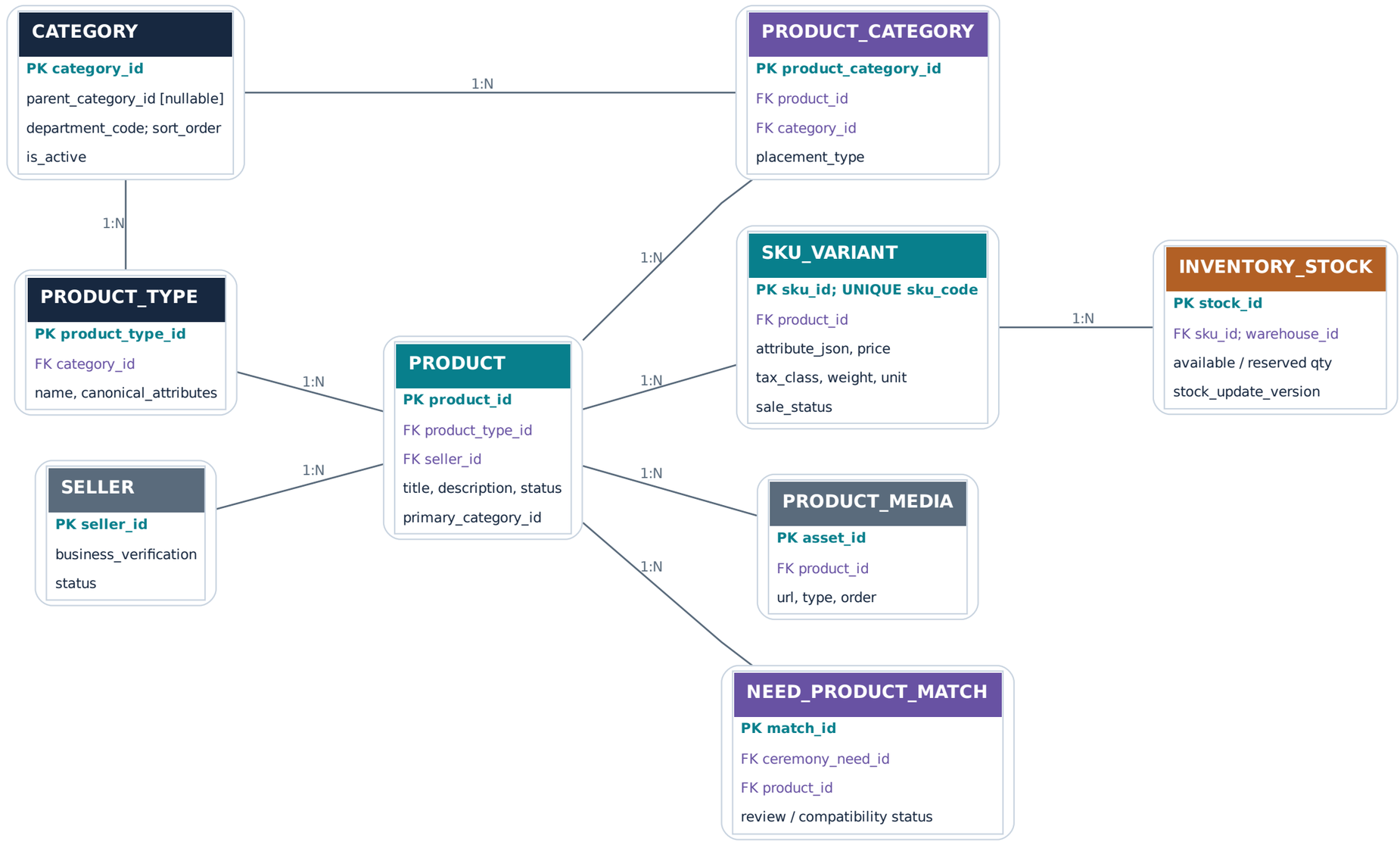
Region / tradition is many-to-many; ceremony requirements are reviewed independently.



The template-ceremony join stores local labels, event order, optionality, review state and evidence.

ER: Products, categories & inventory

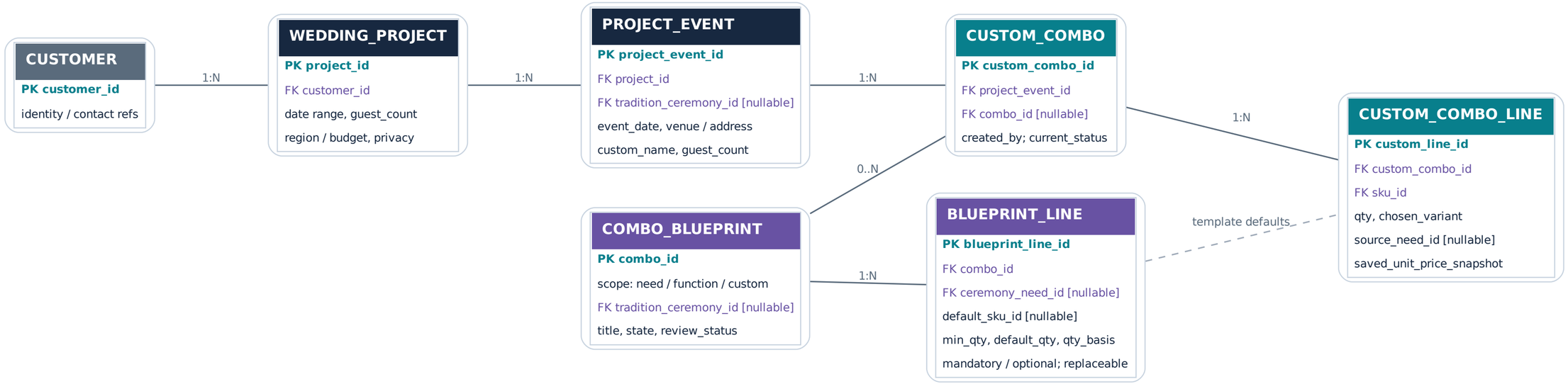
Stable product type and seller product entities prevent duplicate SKUs and stock.



Product-category and product-need mappings are joins. The SKU is the stock and sale unit.

ER: Wedding projects & combos

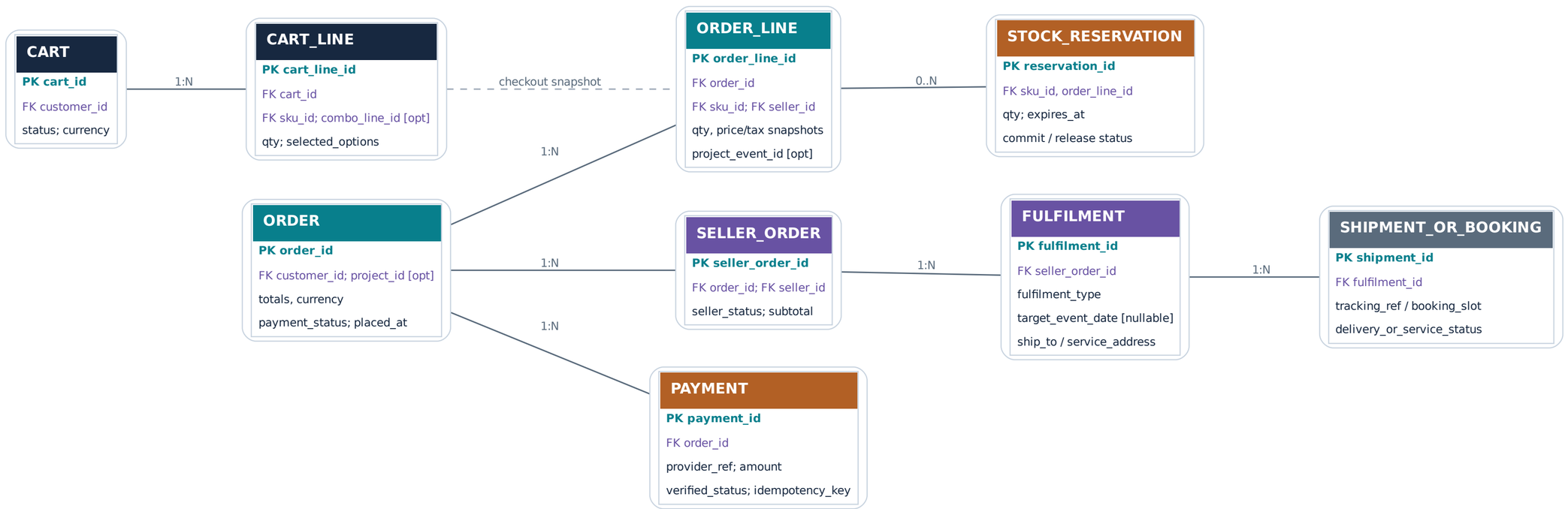
Reusable combo blueprint is independent from the buyer-selected SKU composition.



Guest-count rules affect favours; religion-specific ritual quantities require family or officiant confirmation.

ER: Payments, orders & fulfilment

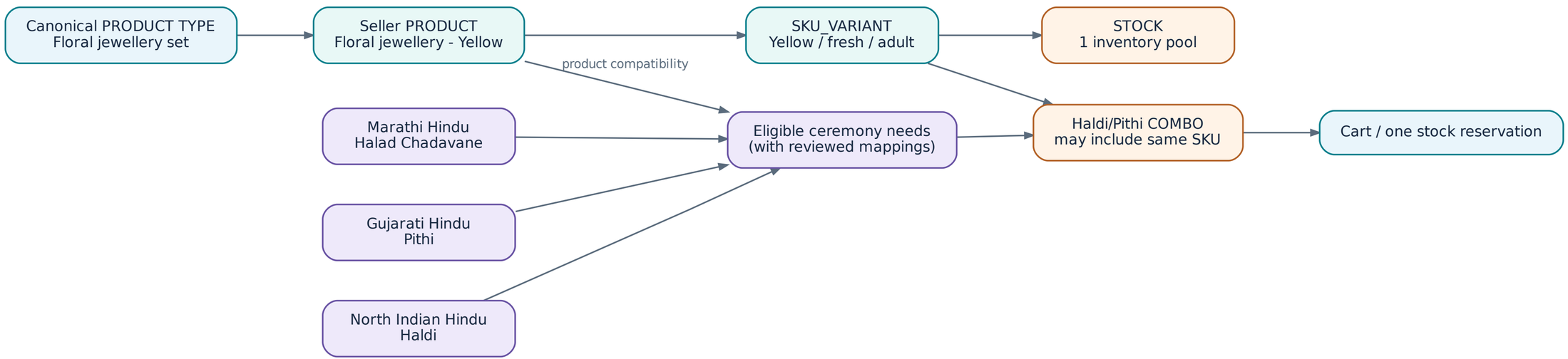
One buyer order may split into multiple seller orders, bookings and deliveries.



Use immutable order-line price snapshots and atomic stock reservations for physical goods.

Example: One SKU in 3 traditions

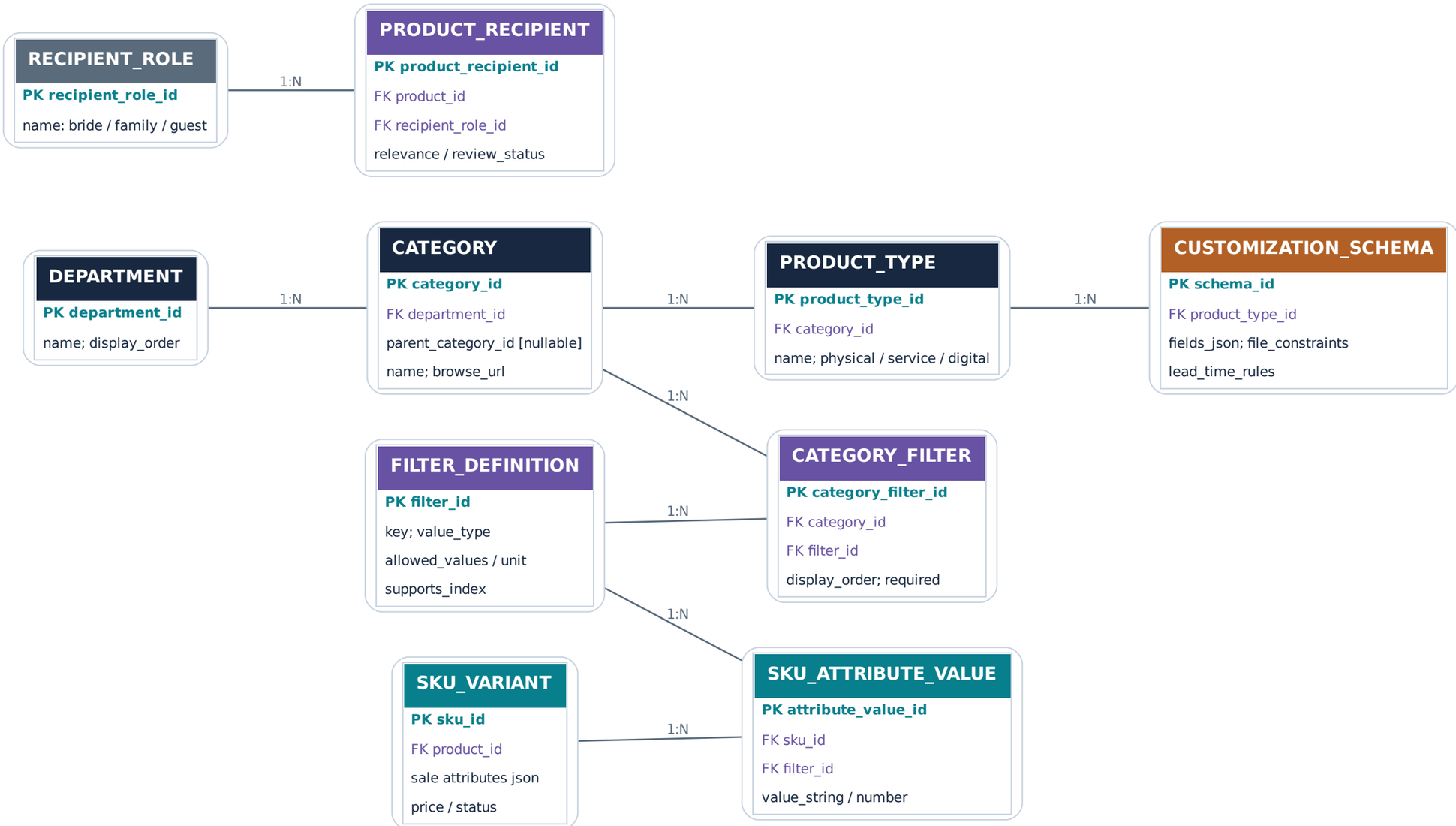
A floral jewellery SKU may be appropriate for multiple reviewed ceremony product needs.



One canonical product and one SKU stock pool serve Marathi Halad, Gujarati Pithi and North Indian Haldi.

ER: Filters, recipients & customization

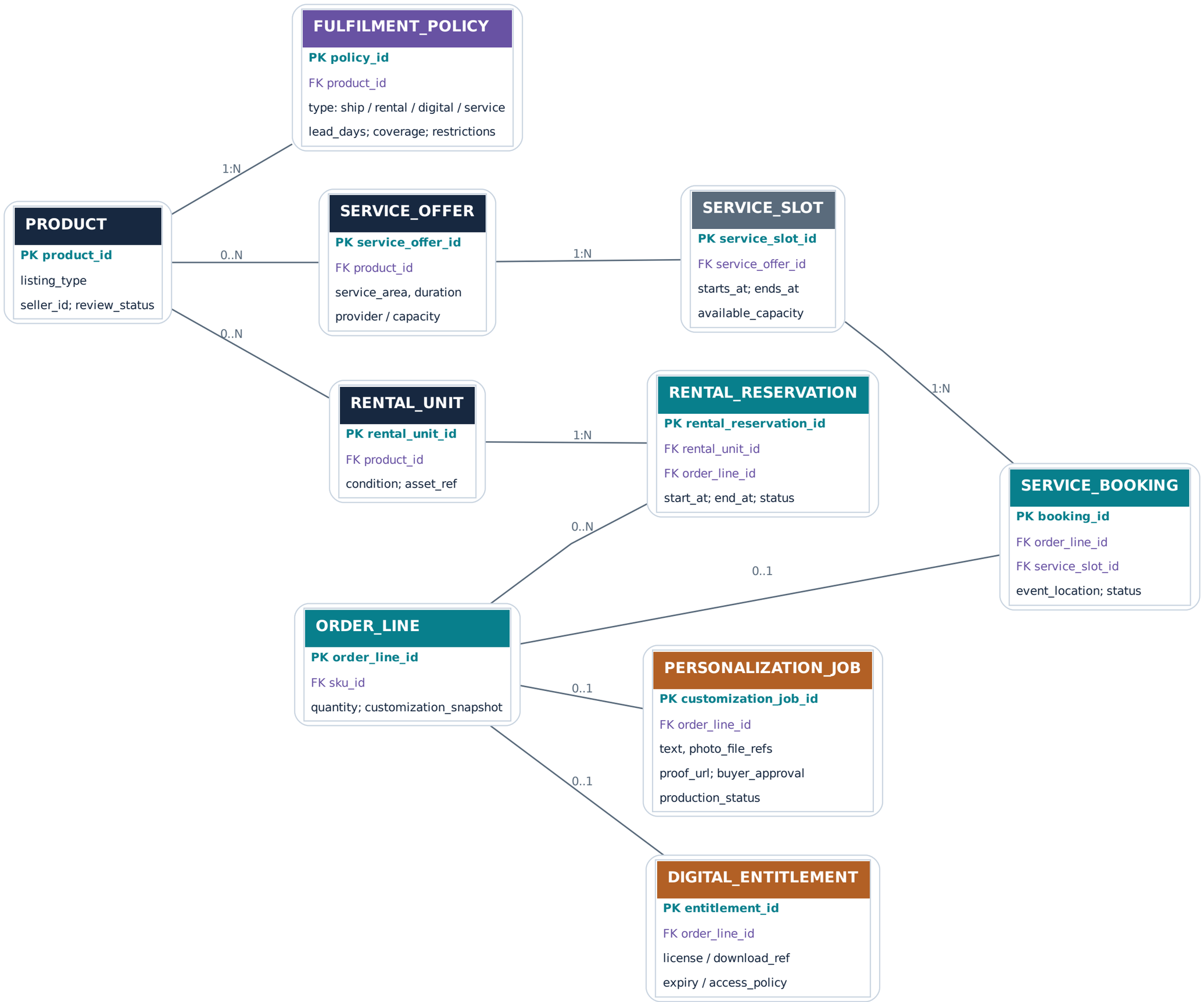
Category filters and SKU values power faceted shopping; recipient roles are separate searchable placements.



Category-defined attributes support size, material, colour and guest count; personalization schemas govern photos and text.

ER: Services, rentals & digital products

Event services, rental inventory, customization proofs and digital downloads require separate fulfilment records.



Bookable service slots and rental periods are capacity-reserved; digital entitlements are not physical SKUs.

Implementation blueprint

Logical data entities can be stored in relational tables or Firestore collections with stable IDs.

Product identity

Category > Product type > Product > SKU variant > Stock by location

Integrity

Unique SKU codes; one authoritative quantity ledger; prevent duplicate reservations.

Wedding identity

Region <-> Tradition <-> Ceremony template <-> Product needs

Cultural safety

Optional family-specific rites and material lists; local reviewer/source status.

Product eligibility

Ceremony need <-> reviewed eligible seller products

Combo safety

Blueprint rules, optional substitutions, selected variant and quantity snapshots.

Buyer plan

Wedding project > selected events > custom combo lines

Commerce

Server-side payments, price/stock checks, correct vendor ownership and taxes.

Checkout

Cart > order lines > seller order > fulfilment & payment

Read models

Firestore browse index can be denormalized, but product and inventory stay canonical.

Open the editable Mermaid / Graphviz files to modify entity fields, relationships, shapes and flow nodes.